12/1/2025
Monday

MODULE - 1

Introduction to Cloud Computing

. Cloud computing is a technology that provides various computing resources such as servers, memory, network, etc: over the internet.

* data stores

. We need to "pay per use"
. It needs constant power supply
. 24 x 7 monitoring
. natural disasters } → Traditional concepts

Examples:-

Google Drive, AWS, Microsoft Azure

AWS ↓ 2006 ↓ wider cloud platform

Microsoft Azure ↓ 2010

> Key Features

. On-demand access :- Whenever a resource is needed, it can be accessed through cloud.

. Scalability :- increase/decrease resources

. Internet-based

> Types

① Private cloud

② Public cloud

③ Hybrid cloud

1> Private Cloud

only for one organization

* Characteristics

1. Dedicated to a single organization
2. High security and controls
3. Suitable for sensitive data

* Advantages

1. Better data security
2. Customization

* Drawbacks

1. Expensive
2. Not mobile friendly
3. Limited scalability

* Examples

- Private cloud in banks

> 2. Public Cloud

Shared with multiple users

* Characteristics

1. Open for public use and shared among users
2. managed by service providers
3. Cost - effective

* **Advantages**

① Low cost
② Easy to use
③ No maintenance is required by user
④ High scalability and elasticity

* **Disadvantages**

① Less security

* **Examples**

AWS, Microsoft Azure, Google Drive, Alibaba cloud, IBM cloud, Oracle cloud

7 ③ **Hybrid Cloud**

Combination of public cloud and private cloud.

. Here, sensitive data is stored in private cloud and less critical data runs on public cloud.

* sensitive data → Private
  Non-sensitive → Public

* **Characteristics**

(i) Flexibility
(ii) Security
(iii) Cost-Effective

Lab

*

Install "Oracle Virtual Box"

Homework

· Look for the terminologies :-

(i) jenkins (ii) DevOps (iii) CI / CD

(iv) version control → git (v) docker

(i) Jenkins :- Automation tool used in DevOps

(ii) DevOps :- Development (Dev) + Operations (Op)

Improve collaboration b/w developers and system admins → planning, development, testing, deployment

(iii) CI / CD :-

CI → Continuous Integration

CD → Continuous Delivery / Deployment

(iv) Git :- Version control system

→ keeps track of code changes

→ Allows multiple developers to work together

→ We can go back to older versions

(v) Docker :- Containerization tool

→ Packages our application with all its dependencies

→ Runs the app the same way on every system

(vi) Container :- Lightweight environment for running apps.

DevOps and cloud } DevOps uses cloud infrastructure to automate building, testing and deploying application efficiently

16/11 Friday

Difference b/w Public cloud and Private cloud

Public cloud	Private cloud
· owned & managed by 3rd party cloud providers	· owned & managed by single organization
· Shared by multiple organization	· Shared by single organization
· Low cost	· High cost
· Low security	· High security

> Multi - Cloud

It uses 2 or more cloud service providers from a number of different computing vendors.

* Purpose :- Reduce risks

· Cloud platforms provide a variety of services and resources over the internet, allowing users ~~and~~ access and utilize computing power, storage, db, networking,

* Types difference
* Types services

* What is virtual box
* installation steps
* gcc compiler
* C program.

16/1/2026 Friday

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* Disaster recovery and backup solutions :

backup and recovers the ...

> Cloud Services

computing services delivered over the internet.

Importance :- agility (ability to move quickly)

· scalability

· cost - effectiveness

① IaaS (Infrastructure as a service)

② PaaS (Platform as a service)

③ SaaS (Software as a service)

> IaaS :- provide virtualized computing resources over the internet.

* servers, storage & networking infrastructure

* Benefits :-
scalability, flexibility, cost - effectiveness

Eg :- AWS EC2, Microsoft Azure Virtual Machines, Google Compute Engine

instance :- allows users to rent virtual servers.

· used to host website & web applications

· for running s/w applications

Uses of IaaS

(1) **Development and Testing environments**

used to create scalable develop

hosting dynamic websites

(2) **Web hosting and application hosting**

ആദ്യം host ചെയ്യണം Web ചെയ്യണം finally ഇവിടെ host ചെയ്യുന്നു.

(3) **Disaster recovery and backup solutions**

disaster വന്നാൽ, data backup ചെയ്യണം, system recover ചെയ്യുന്നു.

(4) **Virtual Data Centers**

AWS EC2 it can replace physical vendors with cloud infrastructure.

> **PaaS**